

1. Administrative & Study Identification

Full Study Title: A Phase Ib/II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating The Efficacy and Safety of Multiple Treatment Combinations in Patients With Locally Advanced Squamous Cell Carcinoma of the Head and Neck (Morpheus-Head and Neck Cancer) **Short Title/Acronym:** Morpheus-Head and Neck Cancer **Protocol Number:** C043613 **NCT Number:** NCT05459129 **Sponsor Name:** Roche **Investigational Product:** Atezolizumab, Tiragolumab, Carboplatin, Paclitaxel, and SBRT **Phase of Development:** PHASE1, PHASE2 **Trial Status:** COMPLETED **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of multiple treatment combinations based on the percentage of participants with pathologic complete response (pCR) at the time of surgical resection, as determined by local pathologic review. **Secondary Objectives:** To evaluate efficacy based on Pathologic Response Rate (pRR) determined by local pathologic review, Event-Free Survival (EFS), and Relapse-Free Survival (RFS), and to assess the safety and tolerability of the treatment regimens.

2.2 Background & Rationale To accelerate the development of treatments or treatment combinations for patients with resectable, locally advanced squamous cell carcinoma of the head and neck (SCCHN). The Morpheus umbrella trial is designed to identify early efficacy signals and establish proof-of-concept clinical data by assessing the importance of simultaneously targeting multiple mechanisms of immune escape (through immune cell priming, activation, and tumor infiltration) in the neoadjuvant setting prior to surgical resection.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Study) **Control Type:** Active Comparator / Multiple Experimental Arms (Atezolizumab monotherapy vs. Atezolizumab + Tiragolumab vs. Combinations with SBRT or Chemotherapy) **Blinding:** Open-label **Randomization:** Randomized **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Target \$N\$: 12 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 14-Jul-2022 **Estimated Primary Completion:** 15-Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 2. Histologically confirmed, resectable Stage III-IVA SCCHN; eligible candidate for R0 resection with curative intent at the time of screening. 3. HPV-negative test for oropharyngeal carcinoma, as determined locally by p16 immunohistochemistry (IHC), in situ hybridization, or polymerase chain reaction-based assay. 4. Measurable disease (at least one target lesion), as assessed according to RECIST v1.1. 5. PD-L1 expression, defined as a combined positive score (CPS) ≥ 1 . 6. Adequate hematologic and end-organ function. 7. Negative HIV test (patients with a positive HIV test at screening are eligible provided they are stable on anti-retroviral therapy, have a CD4 count $\geq 200/\mu\text{L}$, and have an undetectable viral load). 8. Negative hepatitis B surface antigen (HBsAg) test at screening. Positive hepatitis B surface antibody (HBsAb) test at screening, or negative HBsAb at screening accompanied by either negative total hepatitis B core antibody (HBcAb), or positive total HBcAb followed by a negative quantitative hepatitis B virus (HBV) DNA.

4.2 Exclusion Criteria 1. HPV-positive oropharyngeal cancer or distantly metastasized SCCHN. 2. Any prior therapy for SCCHN, including immunotherapy, chemotherapy, or RT; or prior treatment with any of the protocol-specified study treatments. 3. Treatment with investigational therapy within 42 days prior to initiation of study treatment. 4. Treatment with systemic immunostimulatory agents within 4 weeks (or 5 half-lives), or systemic immunosuppressive medication within 2 weeks prior to initiation. 5. Prior allogeneic stem cell or solid organ transplantation. 6. Treatment with a live, attenuated vaccine within 4 weeks prior to study treatment, or anticipation of need within 5 months after the final dose. 7. Active or history of autoimmune disease or immune deficiency. 8. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT scan. 9. History of malignancy other than SCCHN within 5 years prior to screening (excluding malignancies with a $\geq 90\%$ 5-year OS rate). 10. Active tuberculosis, severe infection within 4 weeks, or therapeutic/prophylactic antibiotics within 2 weeks prior to study treatment. 11. Significant cardiovascular disease (NYHA Class II+, myocardial infarction, or CVA within 3 months, unstable arrhythmia, or unstable angina). 12. Major surgical procedure within 4 weeks prior to initiation, or anticipated need for major surgery other than tumor resection. 13. History of severe allergic reactions to chimeric or humanized antibodies, fusion proteins, or known hypersensitivity to Chinese hamster ovary cell products, recombinant human antibodies, or study drugs/excipients. 14. Pregnancy or breastfeeding, intention of becoming pregnant, eligible only for the control arm, or active/suspected chronic EBV infection at screening. 15. **Specific Exclusion for Atezo+Tira+CP:** Known severe allergy or

hypersensitivity to paclitaxel, platinum or platinum-containing compounds, or products containing Cremophor EL; Creatinine clearance ≤ 45 mL/min.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Treatment is administered in the neoadjuvant setting (2 cycles over 6 weeks prior to surgery). Patients are randomized to one of four arms: * Arm 1: Atezolizumab monotherapy * Arm 2: Atezolizumab + Tiragolumab * Arm 3: Atezolizumab + Tiragolumab + Stereotactic Body Radiotherapy (SBRT) * Arm 4: Atezolizumab + Tiragolumab + Carboplatin/Paclitaxel **Route of Administration:** Intravenous (IV) infusions for immunotherapies and chemotherapy; targeted radiation for SBRT. **Duration of Treatment:** 6 weeks of neoadjuvant therapy, followed by standard-of-care surgical resection and subsequent adjuvant follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for treatment-related toxicities, stabilized anti-retroviral therapy for HIV-positive participants. **Prohibited:** Concurrent experimental therapies, other systemic anti-cancer therapies, or immunomodulatory agents outside the specified protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Screening: Informed Consent, Biopsy/Histology Confirmation, RECIST v1.1 Baseline Imaging, Labs, PD-L1/HPV testing
2	Weeks 1 to 6	Neoadjuvant Tx: Randomization, Administration of Investigational Product (2 cycles), Safety Monitoring
3	Week 6 to 8	Surgery: Surgical Resection (R0 intent), Pathological Evaluation of resected tissue (for pCR)
4	Week 12	Post-Op / EOT: Safety Labs, Post-operative Recovery Assessment, End of Treatment check
5	Up to Disease Progression	Follow-up: Surveillance Scans for EFS/RFS, Survival Tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants With Pathologic Complete Response (pCR). pCR is defined as the absence of any viable primary tumor at the time of surgical resection. **Measurement Method:** Local pathologic review.

7.2 Secondary Endpoints * Pathologic Response Rate (pRR) as Determined by Local Pathologic Review. * Event-Free Survival (EFS). * Relapse-Free Survival (RFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring from informed consent through the end of the follow-up period, recording severity per CTCAE criteria. **Serious Adverse Events (SAEs):** Must be reported to the Sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal sponsor and independent committee evaluating real-time safety signals across the umbrella arms to ensure patient safety in the neoadjuvant setting.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy endpoints, Safety Set for adverse events. **Sample Size Justification:** Target enrollment of 12 participants appropriately powered to detect early signals of pCR differences across multiple treatment arms compared to the monotherapy or standard control. **Statistical Methods:** pCR rate was calculated for each arm, along with the 95% confidence interval (CI) estimated using the Clopper-Pearson method. The 95% CI for difference in rates was estimated using the Wald method with continuity correction. Participants with missing or no pathologic response assessment were classified as non-responders.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to verify source data, regulatory document compliance, and surgical pathology reporting. **Audit Trail:** Secure EDC system with complete audit trails, eCRF locking mechanisms, and robust documentation.

11. Study Identifiers & Governance

Primary ID: C043613 **Registry Number:** NCT05459129 **Sponsor/Lead Organization:** Roche **Study Title:** Morpheus-Head and Neck Cancer **Official Regulatory Title:** A Phase Ib/II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating The Efficacy and Safety of Multiple Treatment Combinations in Patients With Locally Advanced Squamous Cell Carcinoma of the Head and Neck (Morpheus-Head and Neck Cancer) **Trial Status:** COMPLETED

12. Clinical Framework (Head & Neck Specific)

Primary Condition: Locally Advanced Squamous Cell Carcinoma of the Head and Neck (LAHNSCC) **Diagnosis Threshold:** Resectable Stage III-IVA SCCHN, HPV-negative, PD-L1 CPS ≥ 1 **Medical Methodology:** Neoadjuvant Immunotherapy / Targeted combination therapy **Optimization Target:** Pathologic Complete Response (pCR) prior to standard surgical resection

13. Study Procedures & Methodology

Management Pathway: Neoadjuvant systemic therapy followed by curative-intent R0 surgical resection. **Surgical Protocol:** Local surgical guidelines strictly enforcing margins and localized pathologic tissue submission. **Biomarker Protocol:** Baseline biopsies sent for PD-L1 CPS scoring, p16 IHC (for HPV status), and immune escape tracking. **Quality Mechanism:** Local pathologic review to confirm pCR and pRR strictly post-surgery.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until disease progression, unacceptable toxicity, or death. **Onsite Visit Cadence:** Frequent monitoring during the 6-week neoadjuvant phase, followed by surgical hospitalization and immediate post-op checks. **Surveillance Scan Frequency:** Ongoing assessments to evaluate Event-Free Survival (EFS) and Relapse-Free Survival (RFS). **Survival Review Interval:** Ongoing long-term follow-up periodically post-progression.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				